

Project Design Phase

Solution Architecture

Date	7 July 2026
Team ID	SWTID-2026-2886
Project Name	Plugging into the Future: An Exploration of Electricity Consumption Patterns
Maximum Marks	4 Marks

Solution Architecture

The proposed solution is designed to collect, process, analyze, and visualize electricity consumption data from different Indian states. The system uses MySQL as the central database, Tableau for interactive dashboards and stories, and HTML, CSS, and Flask to publish the dashboard as a web application. The architecture ensures that users can easily explore electricity consumption trends, compare data across years, regions, and states, and generate insights for better energy planning and decision-making.

Key Features of the Solution Architecture

- Collects electricity consumption data from structured datasets.
- Stores cleaned and processed data in a MySQL database.
- Connects Tableau Desktop with MySQL for live data analysis.
- Generates interactive dashboards, maps, charts, and stories.
- Allows filtering by Year, Month, Region, and State.
- Supports dynamic Top N and Bottom N analysis using parameters.
- Publishes dashboards through Tableau Public.
- Displays the project on a web interface developed using Flask, HTML, and CSS.
- Helps identify electricity consumption trends before and during the COVID-19 lockdown.
- Provides an easy-to-use and scalable analytics solution for researchers and decision-makers.

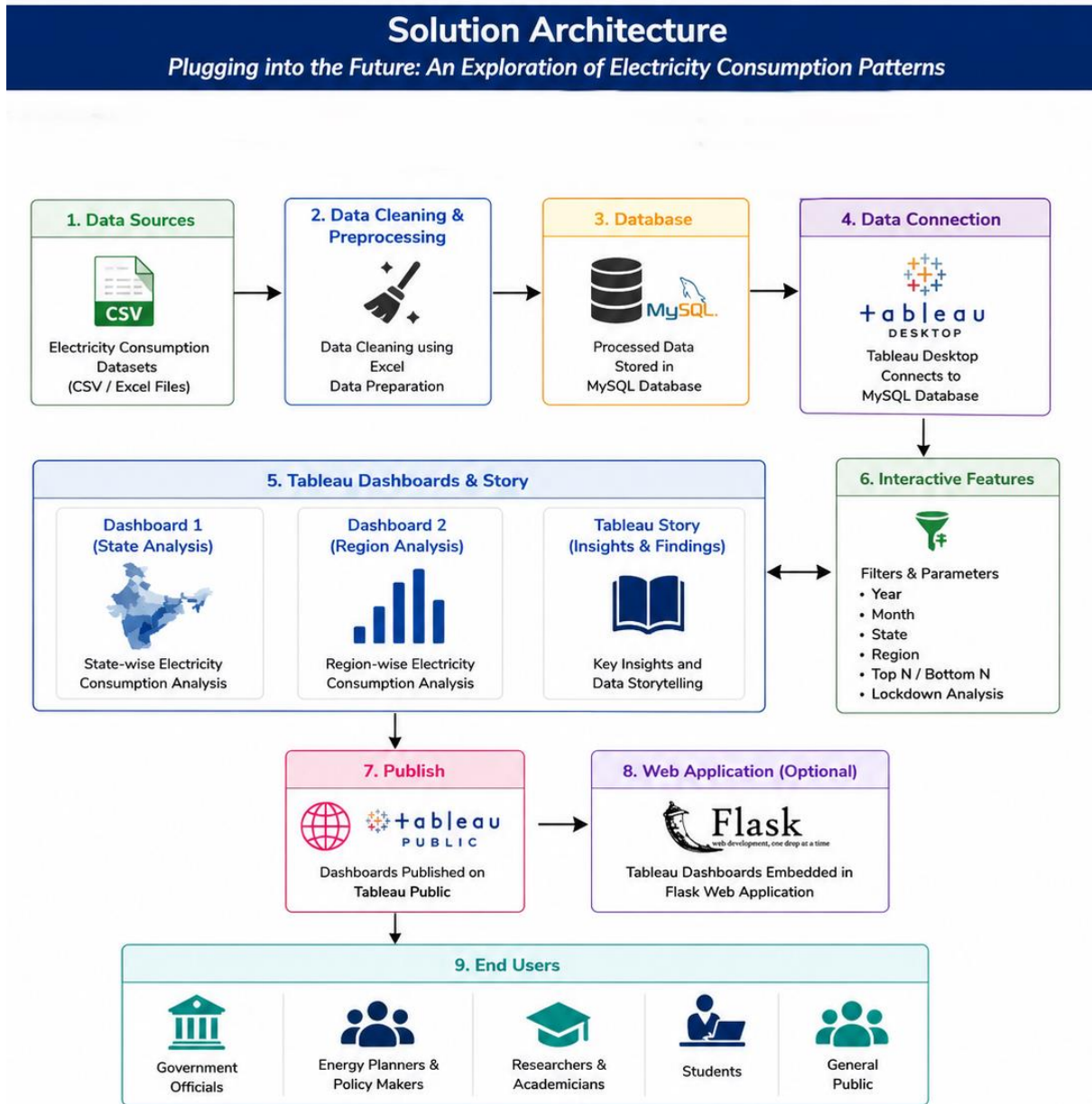


Figure 1: Architecture and data flow of the voice patient diary sample application

Reference: <https://aws.amazon.com/blogs/industries/voice-applications-in-clinical-research-powered-by-ai-on-aws-part-1-architecture-and-design-considerations/>